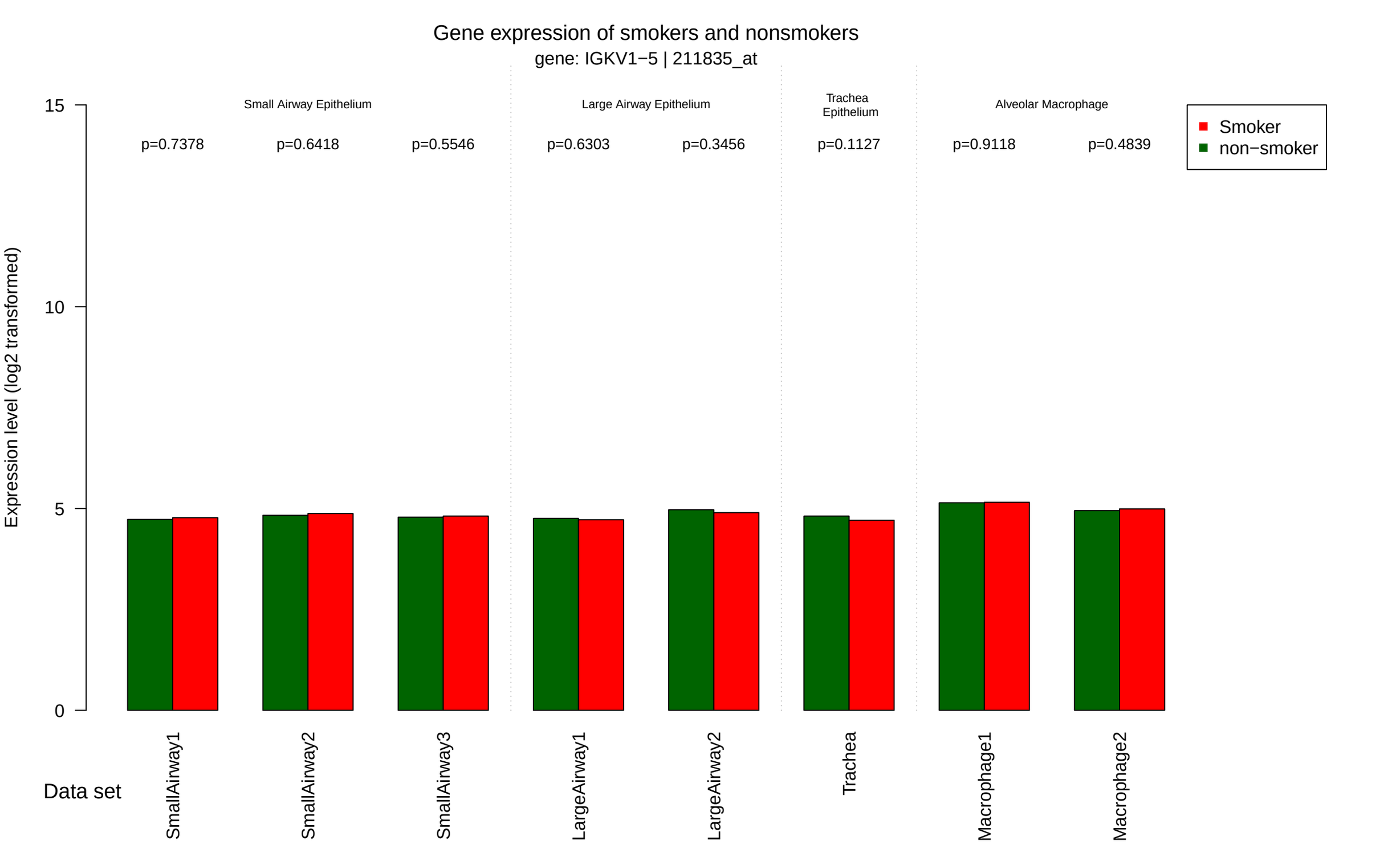
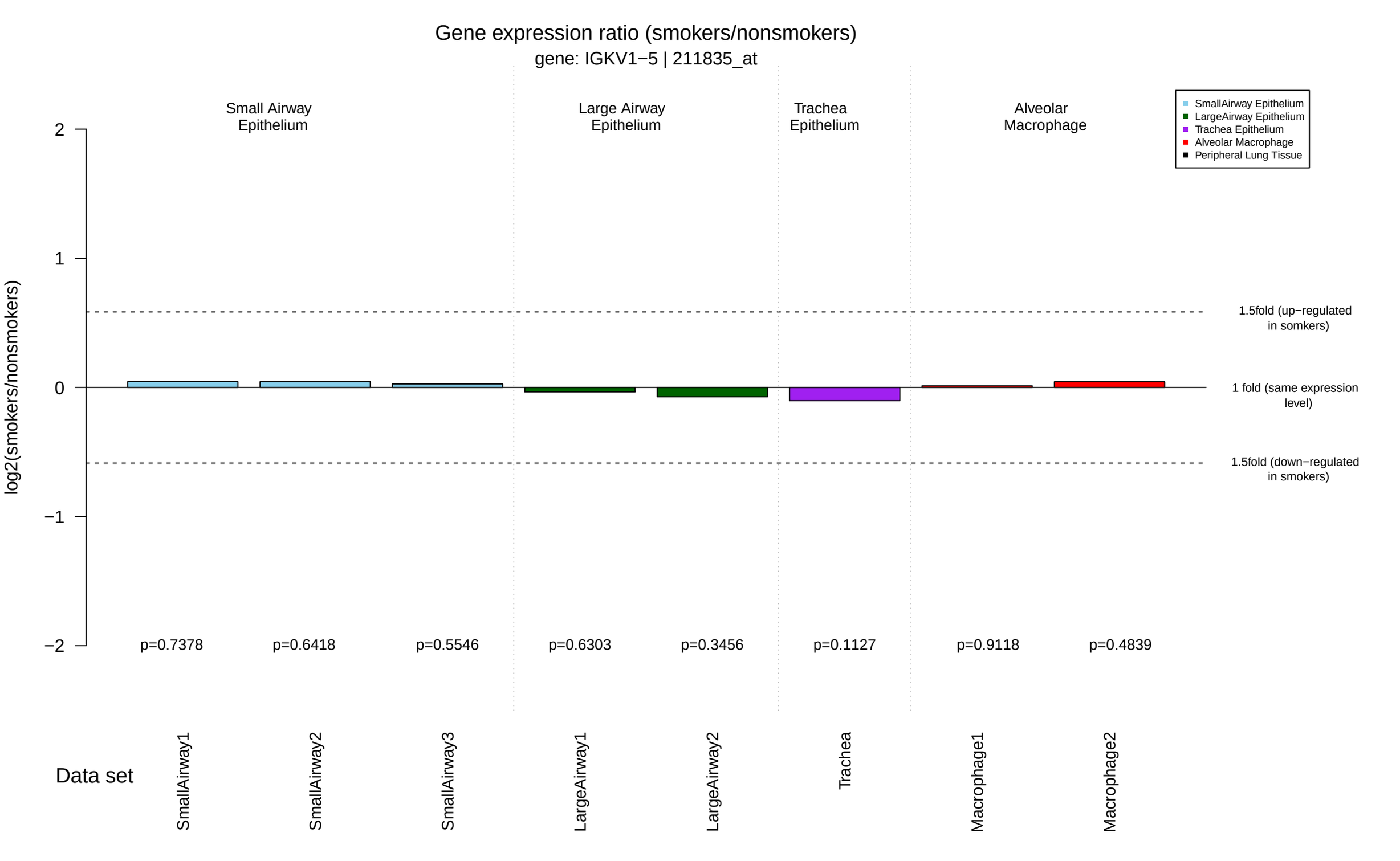
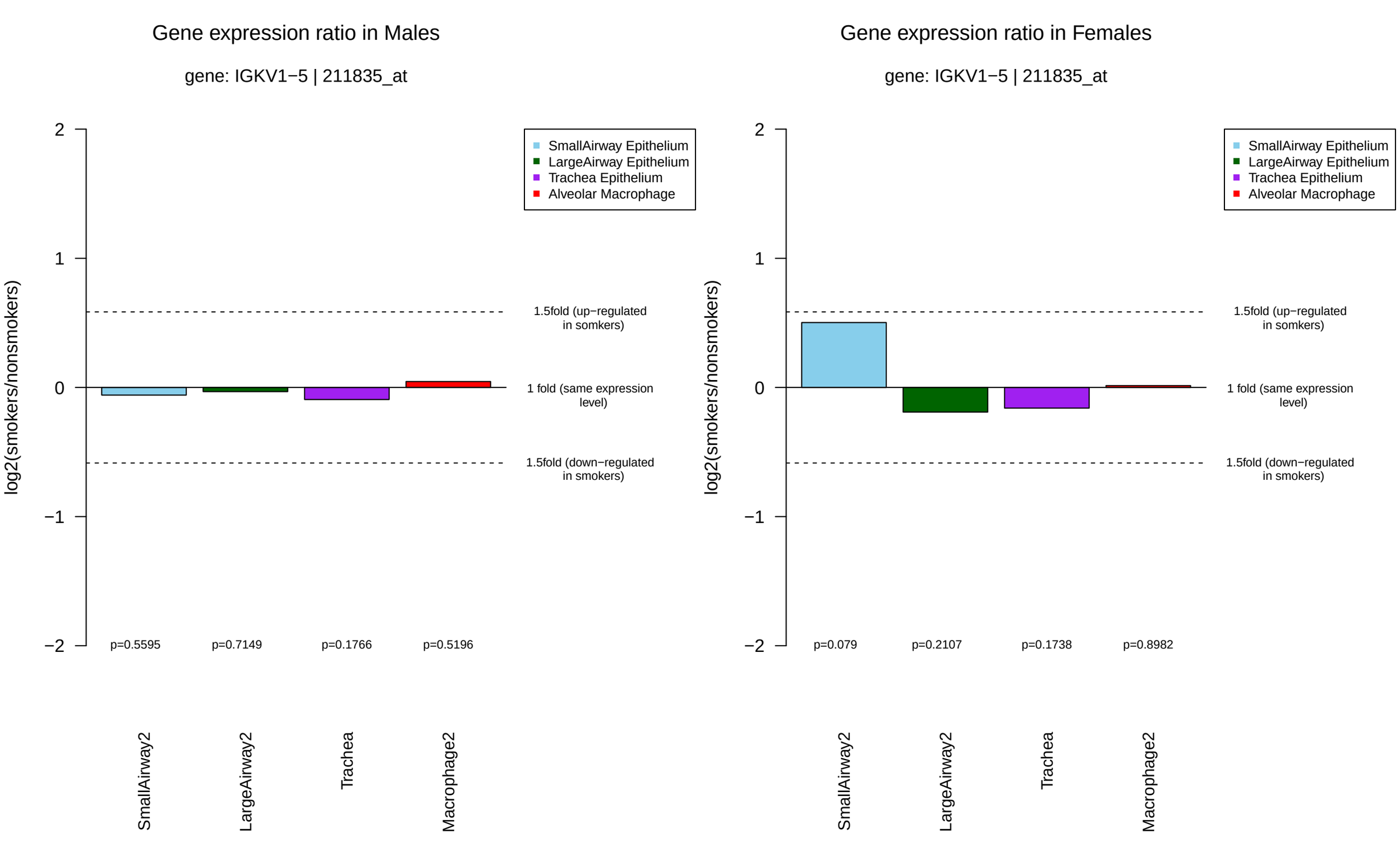
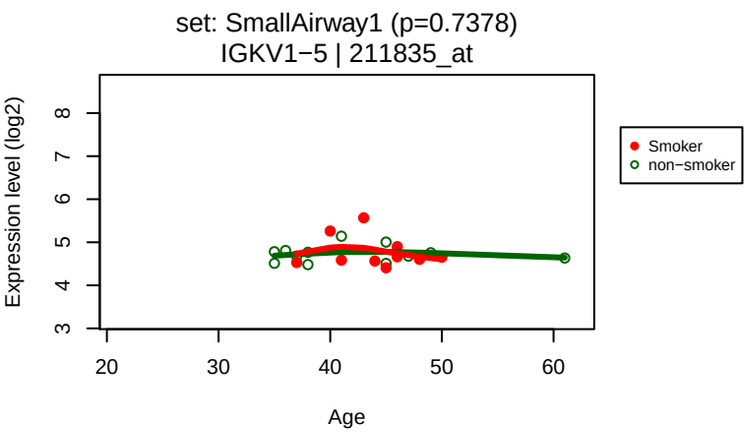
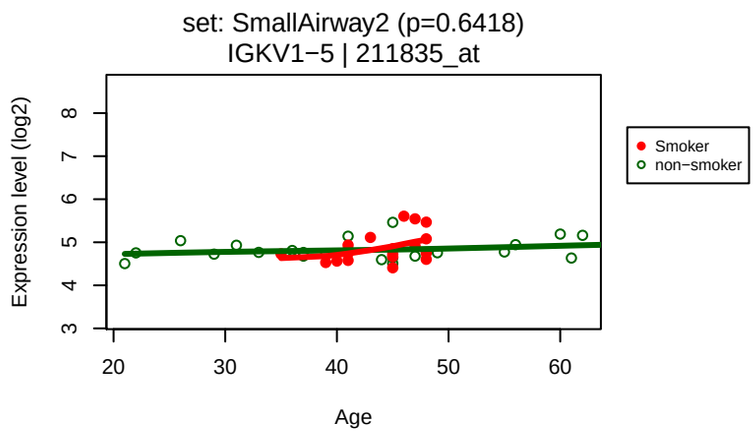
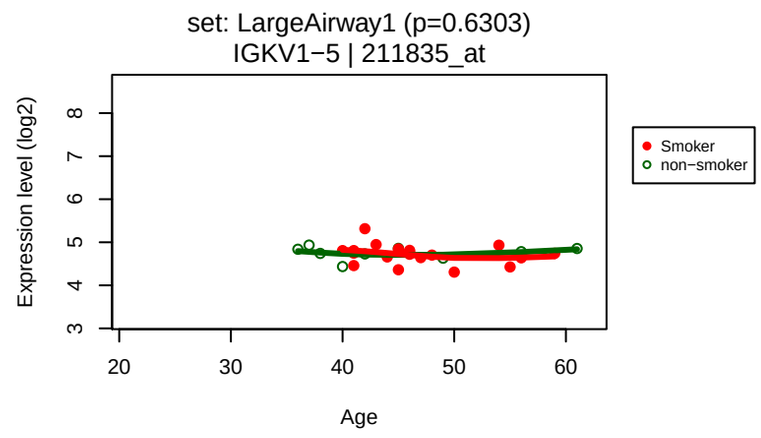

Gene expression of smokers and nonsmokers
gene: IGKV1−5 | 211835_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7378
p=0.6418
p=0.5546
p=0.6303
p=0.3456
p=0.1127
p=0.9118
p=0.4839
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

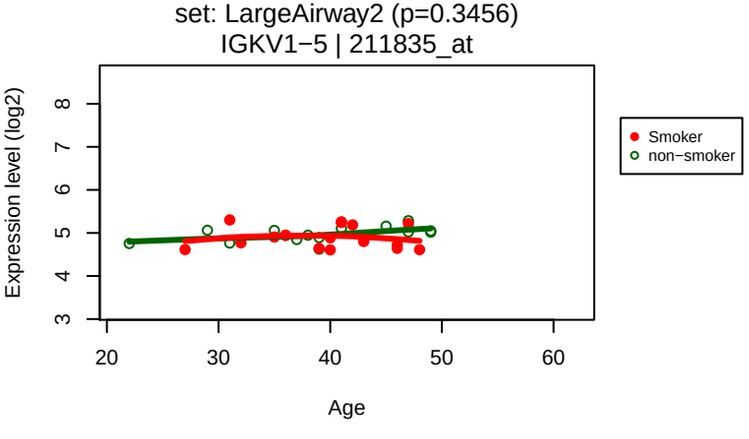

Gene expression ratio (smokers/nonsmokers)
gene: IGKV1−5 | 211835_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7378
p=0.6418
p=0.5546
p=0.6303
p=0.3456
p=0.1127
p=0.9118
p=0.4839
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

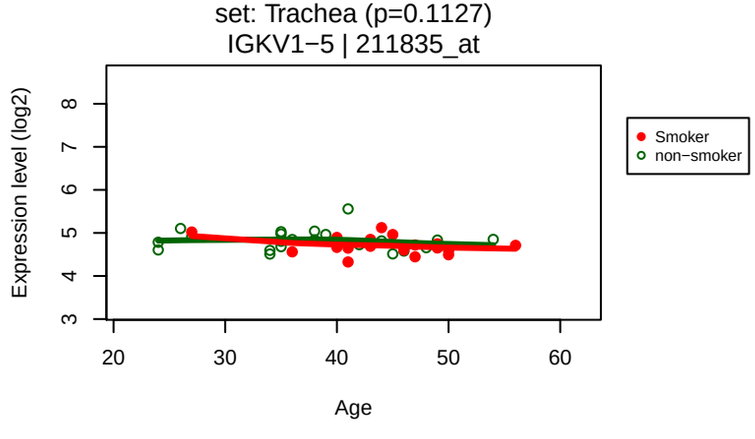

Gene expression ratio in Males
gene: IGKV1−5 | 211835_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5595
p=0.7149
p=0.1766
p=0.5196
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGKV1−5 | 211835_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.079
p=0.2107
p=0.1738
p=0.8982
SmallAirway2
LargeAirway2
Trachea
Macrophage2

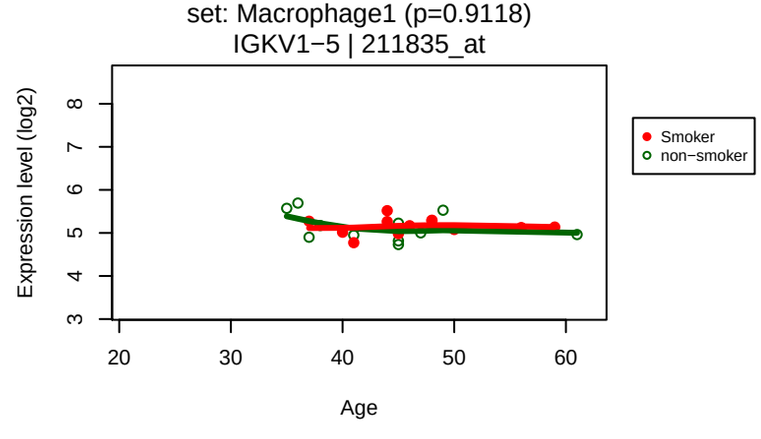

set: SmallAirway1 (p=0.7378)
IGKV1−5 | 211835_at
Expression level (log2)
Age
Smoker
non−smoker

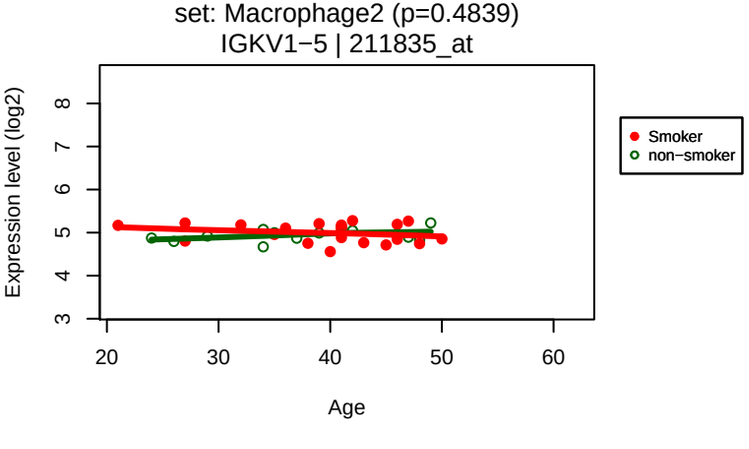

set: SmallAirway2 (p=0.6418)
IGKV1−5 | 211835_at
Expression level (log2)
Age
Smoker
non−smoker

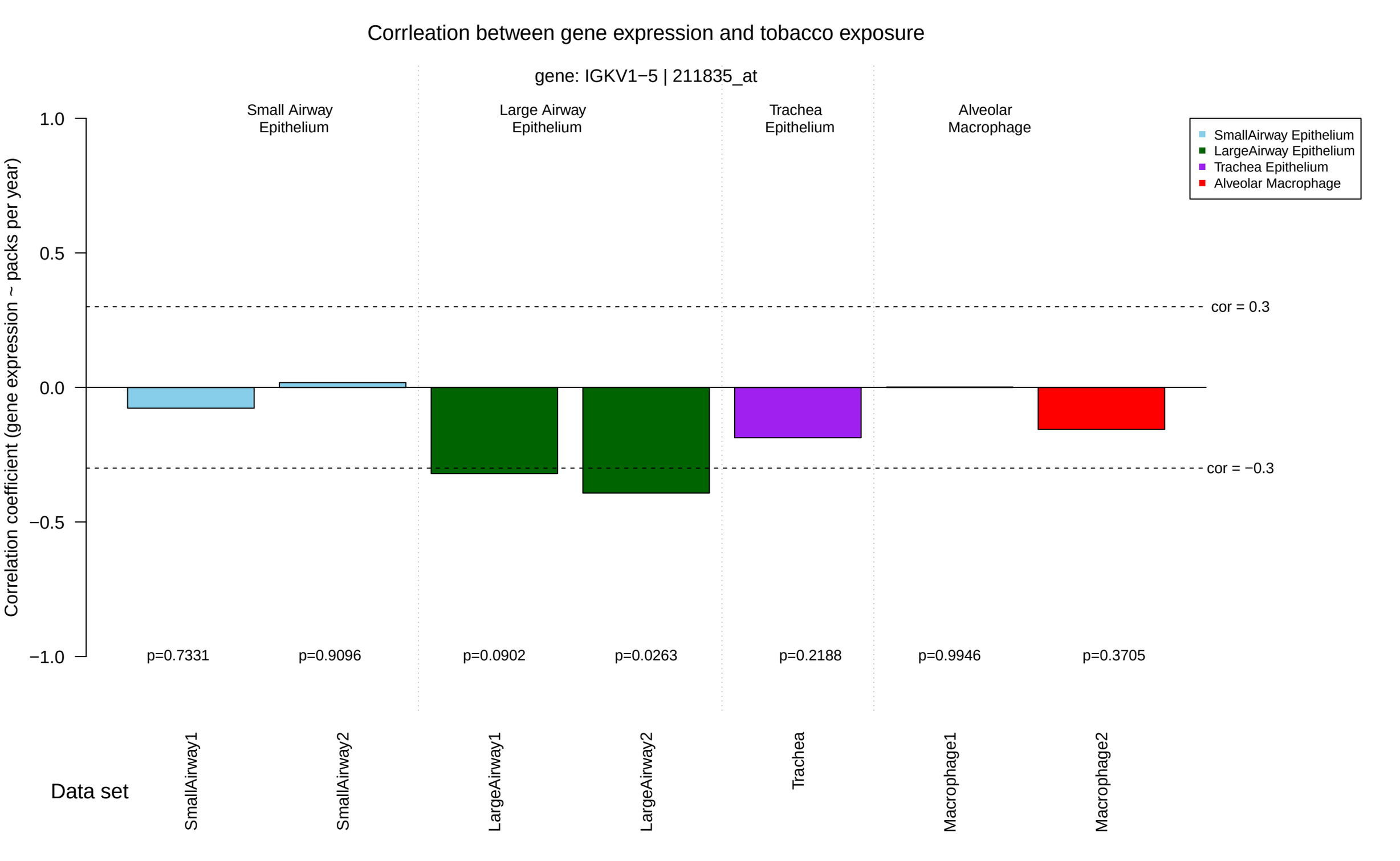

set: LargeAirway1 (p=0.6303)
IGKV1−5 | 211835_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3456)
IGKV1−5 | 211835_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1127)
IGKV1−5 | 211835_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9118)
IGKV1−5 | 211835_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4839)
IGKV1−5 | 211835_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGKV1−5 | 211835_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7331
p=0.9096
p=0.0902
p=0.0263
p=0.2188
p=0.9946
p=0.3705
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2